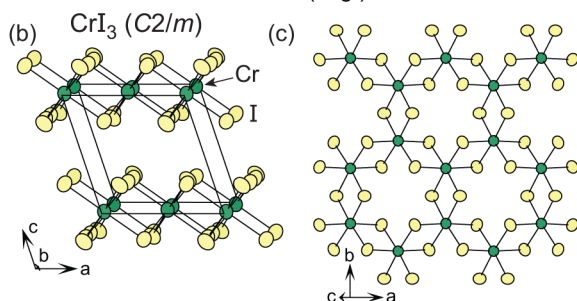
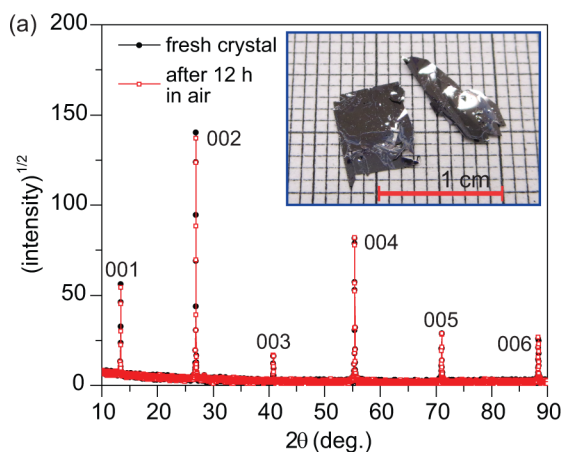


1

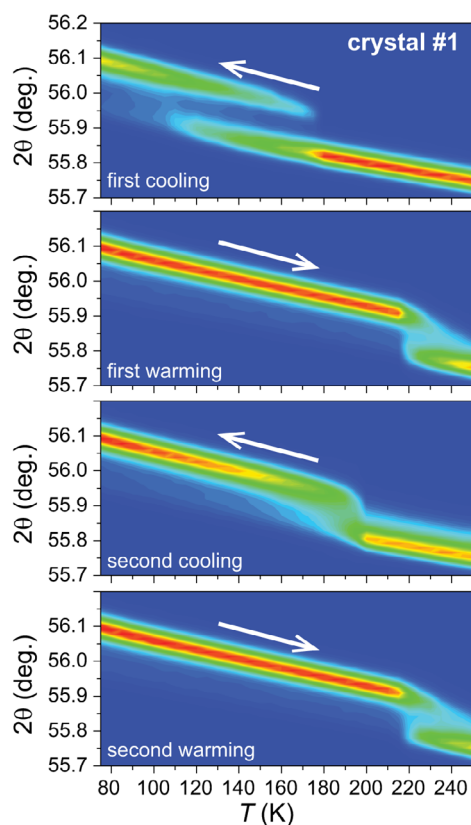


3

T (K)	250	90
space group	$C2/m$	$R\bar{3}^a$
a (Å)	6.866(3)	6.867(4)
b (Å)	11.886(5)	6.867(4)
c (Å)	6.984(3)	19.807(17)
β (deg)	108.507(7)	90
vol (Å ³)	540.5(4)	809(1)
Cr1 position	0, 0.16654(5), 0	$1/3, 2/3, 0.33299(7)$
I1 position	0.22407(5), 0, 0.23862(4)	0.31677(9), 0.33453(5), 0.41230(3)
I2 position	0.25110(3), 0.32452(2), 0.23467(4)	
R1/wR2, all data	0.020/0.049	0.057 [0.031] ^b /0.123 [0.069] ^b
Cr–I distances (Å)	2.7267(8), 2.7185(8), 2.7196(8)	2.725(2), 2.727(2)
Cr–Cr distances (Å)	3.959(2), 3.965(1)	3.965(2), 3.965(3)
I–I intralayer short distances (Å)	3.852(1), 3.862(1), 3.863(1)	3.857(2), 3.878(2)
I–I intralayer long distances (Å)	4.172(2), 4.184(1)	4.168(3)
I–I interlayer avg distances (Å)	4.155(1)	4.160(3)
layer spacing (Å)	6.623	6.602
vdW gap (Å)	3.488	3.471

^aReverse-obverse twinning present with a twin fraction of 0.60.^bValues in square brackets are for the model including a small amount of disorder described in the text and the SI.

2



4

